II.2314 / II.2414 - Advanced Databases and Big Data

Flight Fare Analytics and Fuel Price Integration Using Big Data Technologies

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Dashboard Link:
http://3.93.194.228:5601/app/r/s/ByRqg

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Abstract:

The airline industry generates a large volume of data related to flight operations, ticket pricing, and customer travel patterns. Analyzing this data can help identify pricing trends and support business decision-making. This project presents a Big Data analytics solution that combines flight fare data with fuel price information to study airfare patterns and provide meaningful insights.

The solution was developed using Python, Apache Airflow, Elasticsearch, and Kibana. The flight dataset was processed and enriched with fuel price information obtained from an external source. The processed data was then loaded into Elasticsearch and visualized using Kibana dashboards. The project demonstrates how modern Big Data technologies can be used to collect, process, store, and analyze large-scale aviation data efficiently.

Introduction:

Air ticket pricing is influenced by several factors such as airline operators, travel routes, booking dates, travel class, flight duration, and operational costs. Fuel cost is one of the most important expenses for airlines and can significantly impact ticket prices.

The main objective of this project is to build a complete Big Data pipeline that integrates flight fare information with fuel price data and provides interactive visual analytics for understanding airfare trends.

The project focuses on:

- Data ingestion and processing
- Integration of multiple data sources
- Data storage using Elasticsearch
- Dashboard creation using Kibana
- Exploratory analysis of flight pricing trends

Problem Statement:

Airline ticket prices vary significantly depending on multiple factors. Traditional data analysis approaches are often insufficient when dealing with large datasets and multiple data sources.

The problem addressed in this project is:

"How can Big Data technologies be used to analyze flight ticket prices and integrate external fuel price information to generate meaningful business insights?"

Objectives:

The primary objectives of this project are:

- Analyze flight ticket pricing patterns.
- Integrate external fuel price data into the flight dataset.
- Build an automated data processing pipeline.
- Store processed data in Elasticsearch.
- Create interactive dashboards using Kibana.
- Generate insights regarding airline pricing strategies.
- Explore the possibility of predictive analytics using machine learning.

Dataset Description:

a) Flight Price Dataset

The flight dataset contains more than 300,000 flight records with information such as:

- Airline
- Flight Number
- Source City
- Destination City
- Departure Time
- Arrival Time

- Number of Stops
- Travel Class
- Duration
- Days Left Before Departure
- Ticket Price

b) Fuel Price Dataset:

A fuel price dataset was integrated into the project as an external economic indicator. The latest fuel price was collected and merged with the flight dataset during the ETL process.

Attributes:

- Date
- Fuel Price

Technologies Used:

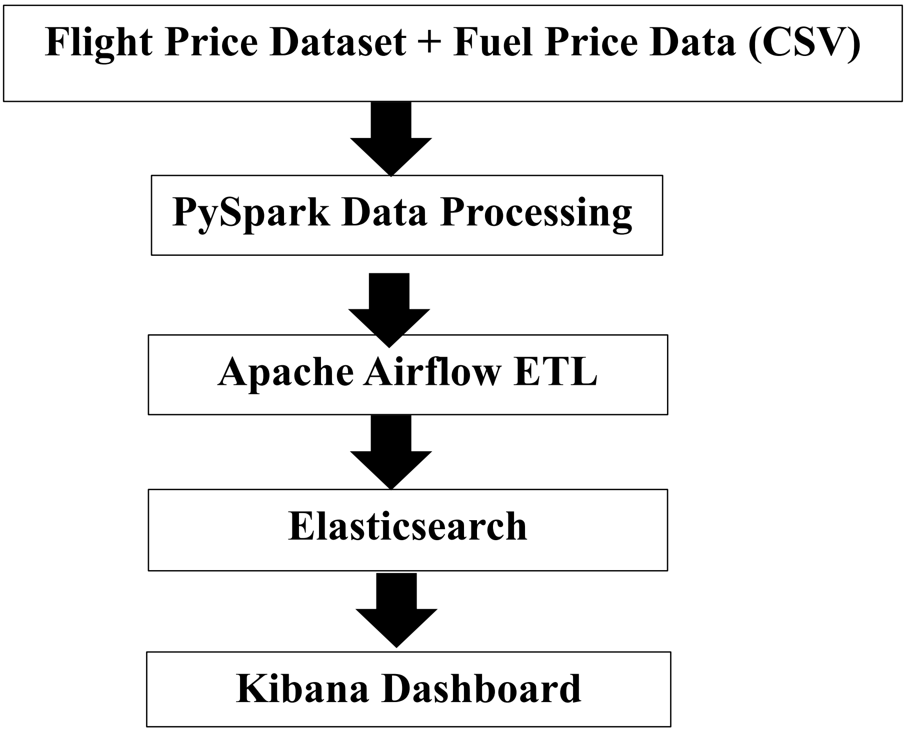
The following technologies were used in this project:

Technology	Purpose
Python	ETL Development
PySpark	Big Data Processing
Apache Airflow	Workflow Automation
Elasticsearch	Data Storage & Search Engine
Kibana	Data Visualization

Docker	Containerization
Scikit-Learn	Machine Learning
GitHub	Version Control

System Architecture:

The project follows the architecture shown below:



Data Processing Pipeline:

The ETL pipeline consists of three major stages:

Extract

- Flight data is loaded from the CSV dataset.
- Fuel price information is collected from an external source.

Transform

- Data cleaning is performed.
- Unnecessary columns are removed.
- Fuel price information is added to flight records.
- The final integrated dataset is generated.

Load

- Processed records are loaded into Elasticsearch.
- Elasticsearch indexes the data for efficient searching and analytics.

Elasticsearch Integration:

Elasticsearch was used as the primary storage engine for analytical querying.

A dedicated index named:

flight_fuel_data

was created to store processed records.

Benefits:

- Fast search performance
- Scalability
- Real-time analytics support
- Easy integration with Kibana

More than 300,000 flight records were successfully indexed into Elasticsearch.

Kibana Dashboard:

A comprehensive dashboard was developed in Kibana to visualize airfare trends.

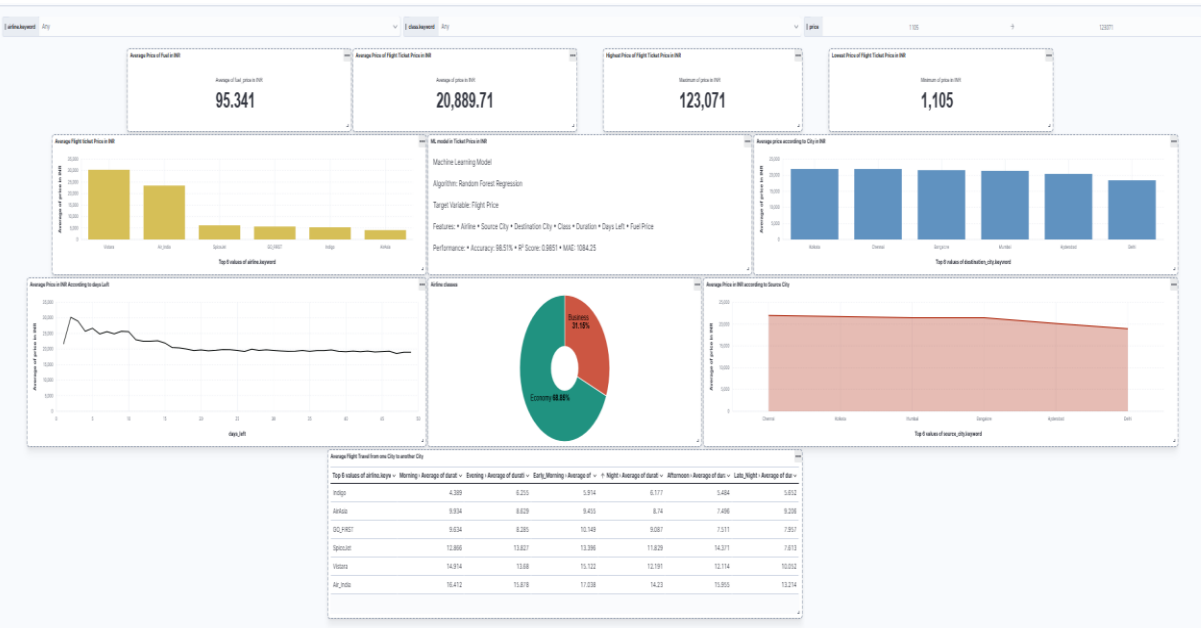
The dashboard includes:

KPI Cards

- Average Fuel Price
- Total Flights
- Average Flight Price
- Maximum Flight Price
- Minimum Flight Price

Visualizations

- Average Flight Price by Airline
- Average Price by Source City
- Average Price by Destination City
- Flight Class Distribution
- Price Trend by Days Left
- Average Flight Duration by Airline



Key Findings:

The analysis revealed several important insights:

- Vistara and Air India have higher average ticket prices compared to low-cost carriers.
- Ticket prices tend to increase as the departure date approaches.
- Business-class tickets are significantly more expensive than economy-class tickets.
- Different cities exhibit different pricing patterns.
- Fuel price information can be incorporated into airfare analytics for future cost analysis.

Machine Learning Extension:

To enhance the project, a Machine Learning component was explored using Random Forest Regression.

The objective was to predict flight ticket prices based on:

- Airline
- Source City
- Destination City
- Travel Class
- Duration
- Days Left
- Fuel Price

This extension transforms the project from descriptive analytics to predictive analytics.

Machine Learning Module

Algorithm: Random Forest Regression

Target Variable: • Flight Price

Input Features: • Airline • Source City • Destination City • Travel Class • Duration • Days Left • Fuel Price

Model Performance: • Accuracy: 98.51% • R^2 Score: 0.9851 • MAE: 1084.25

Challenges Faced:

Several challenges were encountered during development:

- Integration of external fuel price data.
- Elasticsearch indexing of large datasets.
- Airflow compatibility issues on Windows.
- Managing large-scale data processing efficiently.

These challenges were resolved through Docker-based deployment and optimized ETL workflows.

Conclusion:

This project successfully demonstrates the application of Big Data technologies for airline fare analysis. Flight pricing data was integrated with fuel price information, processed using Python, stored in Elasticsearch, and visualized through Kibana dashboards.

The developed solution provides valuable insights into airline pricing behavior and demonstrates the practical use of modern data engineering and analytics tools. The project can be further enhanced by incorporating historical fuel prices, real-time data streaming, and advanced machine learning models for improved price prediction.

Future Scope:

1. Real-time flight fare monitoring.
2. Historical fuel price analysis.
3. Streaming analytics using Apache Kafka.
4. Advanced machine learning models.
5. Real-time predictive dashboards.

Links:

Dashboard Link:

http://3.93.194.228:5601/app/r/s/ByRqg

Blog Link:

https://medium.com/@aakashraina2002/flight-fare-analytics-and-fuel-price-integration-using-big-data-technologies-6aea2be09b03

GitHub Link:

https://github.com/Aakash-bit/FlightPriceX-Analytics-Platform

Data Source 1:

https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction

Data Source 2:

Website:
https://www.exchangerate-api.com/

API Documentation:
https://www.exchangerate-api.com/docs